



Fundamental equation(s) of string theory?

Asked 13 years, 2 months ago Modified 10 years, 11 months ago Viewed 36k times



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I often hear about [string theory](#) and its complicated mathematical structure as a physical theory, but I can't say that I've ever actually seen any of the related math. In general, I'm curious as to what the mathematics of string theory look like, can anyone point me to some references? In specific, I want to know if there is a fundamental equation in string theory that is assumed as a starting point for most problems, something comparable to Newton's second law in mechanics or the Schrodinger equation in QM?

string-theory

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edited Apr 24, 2013 at 14:37



Qmechanic ♦

228k ● 52 ● 653 ● 2.7k

asked Apr 24, 2013 at 14:23



Sigma

315 ● 1 ● 2 ● 9

If you like this question you may also enjoy reading [this](#) and [this](#) Phys.SE post. – Qmechanic ♦ Apr 24, 2013 at 14:40

2 Answers

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I've long been interested in this, but the impression I get is (speaking as a strict amateur with a reasonable understanding of QM and relativity) there is simply nothing like e.g. the Schrodinger equation or Einstein's field equation in string theory. String theory is developed by writing down the action (which is the area of the string world sheet), using this to find the (classical) equations of motion, trying to find a consistent quantisation of these (building in supersymmetry somewhere along the way) then solving the resulting impossibly messy and hard equations using perturbation theory. The impression I get (NB as an outsider) is that because it's so hard people have attacked it from many different angles in many different ways so what we know as string theory is really lots of overlapping bits rather than an elegant monolith like GR.

The best non-non-nerd introduction I've read is [String Theory Demystified](#) by David McMahon. If you work through this you can at least get an idea of how it's all put together, though it will still leave you (and me!) far short of anyone who actually works in the field. The Amazon link I've

given allows you to read selected chapters from the book, and in any case it's pretty cheap second hand.

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answered Apr 24, 2013 at 16:19



John Rennie

376k ● 135 ● 810 ● 1.1k

- String theory is formulated using Feynman's sum over history formalism. The basic equation is just the path integral. The thing that makes strings difficult, in some sense, is that we don't understand very well what variables we should be using in this path integral. – [user1504](#) Apr 24, 2013 at 19:16



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What I want to say here is related to user1504's comment.

As Lenny Susskind explains in [this](#) and [this](#) lecture, how to describe the scattering behavior of particles is nearly the definition of string theory. So formulas for scattering amplitudes can in some way be considered as fundamental equations defining the theory. Very schematically, the equation to calculate the scattering amplitude A can be written down as

$$A = \int_{\text{period}} d\tau \int_{\text{surfaces}} \exp^{-iS} \Delta X^\mu(\sigma, \tau)$$

Considering for example the process of two strings joining and splitting again, one has to integrate over all world sheets $\Delta X^\mu(\sigma, \tau)$ that start and end with two distinct strings. A second integral has to be done over all possible periods of time $d\tau$ the strings join. The action S may for example be given by

$$S = \int d\tau d\sigma \left[\left(\frac{\partial X^\nu}{\partial \tau} \right)^2 - \left(\frac{\partial X^\nu}{\partial \sigma} \right)^2 \right]$$

The information about the incoming and outgoing particles themselves is still missing in the first equation and has to be inserted by hand by including additional multiplicative factors (vertex operators)

$$\prod_j e^{ik_{j\mu} X^\mu(z_j)}$$

These factors represent a particle with wave vector k , and z is the location of injection (for example on the unit circle when conformally transforming the problem to the unit disk) over which has finally to be integrated too.

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edited May 12, 2013 at 23:05

answered May 12, 2013 at 22:48

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Dilaton

9,851 ● 3 ● 54 ● 146

The incoming/outgoing particles (vertex operators) are "put in by hand" but naturally so given the state-operator correspondence. – [lionelbrits](#) Nov 4, 2013 at 21:31



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